SHANREIGN FELICIANO

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EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES Los Angeles, CA
Bachelor of Science, Bioengineering Expected June 2029

EXPERIENCE

UCLA Kamariza Lab Los Angeles, CA
Undergraduate Researcher Dec 2025 – Present

- Optimized CRISPR-Cas13a DNA hydrogel diagnostic platform by evaluating hydrogel stability, reporter leakage/diffusion kinetics, and concentration conditions to improve assay performance and reproducibility.
- Developed quantitative Cryo-SEM workflows using Fiji/ImageJ and Labkit to characterize hydrogel pore structure, material morphology, and segmentation reliability.
- Use Fiji/ImageJ and Labkit for image preprocessing, segmentation, morphological analysis, and quantitative characterization of biological materials.
- Analyzed experimental data in GraphPad and Excel to compare conditions, identify sources of variability, and guide subsequent iterations.
- Designed and 3D-printed a custom imaging fixture to standardize sample positioning and photo capture.
- Refined experimental protocols and maintained structured documentation in Benchling to support reproducibility across experiments.

Spartatroniks 3512 - Nonprofit STEM Organization Orcutt, CA
Robotics Mechanical Lead & Robot Captain Oct 2021 – Jun 2025

- Led a 40+ member multidisciplinary engineering team through the complete robot development cycle: CAD, fabrication, mechanical assembly, electrical wiring & design, system integration, testing, troubleshooting, and competition deployment.
- Designed and fabricated mechanical components using SolidWorks, CNC machining, mills, lathes, laser cutters, and FDM 3D printing, translating CAD concepts into functional prototypes.
- Performed mechanical assembly, fit checks, tolerance verification, and iterative redesign to resolve failures discovered during fabrication and testing.
- Implemented quality-control procedures for manufactured components, identifying defects early and coordinating rework that reduced material waste by 15%.
- Developed standardized build logs, parts documentation, and fabrication workflows to improve communication and design continuity across a large engineering team.
- Coordinated rapid troubleshooting of mechanical, electrical, and software failures under competition deadlines, delegating testing and debugging across subteams.

E-NABLE & Adapt3D Orcutt, CA
Hand Prosthetics Volunteer Aug 2024 – Jun 2025

- Designed, fabricated, and assembled 60+ rapid prototypes for custom assistive devices using Blender and FDM 3D printing.
- Iterated device geometry and mechanical interfaces to address structural, fit, and wearability constraints across multiple prototype cycles.
- Evaluated prototypes through hands-on testing and troubleshooting, balancing function, manufacturability, and predicted user fit in design revisions.

ADDITIONAL

- ***Certifications & Training:*** CPR/AED & BLS Certified, CompTIA Tech+ Cert. (May 2025)
- ***Skills*****:** Microsoft Office (Word, Excel, and PowerPoint), SolidWorks, GD&T, mechanical assembly, tolerance/fit checks, rapid prototyping, FDM 3D printing, machining
- ***Analysis*****:** GraphPad, Excel, Fiji/ImageJ, Labkit, quantitative image analysis
- ***Interests*****:** Costco, Tennis/Pickleball, Thrifting, Beach Bonfires, Paintball, Coffee-staining letters, Hiking